

Conjecture 00000001209: the golden-ratio slope fails for 2-Wythoff

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Statement, verbatim (00000001209). “Conjecture: For every $t \geq 1$, the n -th P-position of t -Wythoff is asymptotic to $(\varphi n, \varphi n + tn)$ with an explicit linear correction coefficient (t -Wythoff asymptotics).”

Result. Under the standard definition of the game and the usual meaning of “asymptotic to,” the universal assertion is false. For $t = 2$ the smaller heap has slope $\sqrt{2}$, and

$$\frac{a_n}{\varphi n} \rightarrow \frac{\sqrt{2}}{\varphi} \neq 1, \quad \varphi = \frac{1 + \sqrt{5}}{2}.$$

The accompanying Lean proof establishes the exact P-position characterization from the move relation, rather than assuming Fraenkel’s theorem.

1 Definition and interpretation

We use Fraenkel’s standard t -Wythoff game on two heaps, with an integer $t \geq 1$: a move removes any positive number from one heap, or removes positive numbers k, l from both heaps subject to $|k - l| < t$. Play is normal: a player with no move loses. A P-position is a position from which every move leads to a position that is not a P-position. This is a well-founded recursive definition because every move strictly decreases the sum of the heaps.

Heap order is irrelevant to the winning status. We list the P-positions as (a_n, b_n) with $a_n \leq b_n$, in increasing order of a_n , and include $(0, 0)$ at index 0. Starting the list at 1 makes no difference to any limit here. Coordinatewise asymptotic equivalence requires

$$a_n/(\varphi n) \rightarrow 1, \quad b_n/((\varphi + t)n) \rightarrow 1.$$

It suffices to refute the first requirement for $t = 2$.

Robustness to the non-strict convention. If simultaneous removals instead satisfy $|k - l| \leq t$, integer removal amounts make this exactly the strict game with parameter $t + 1$. Consequently its $t = 1$ instance is already not classical Wythoff: the first positive sorted P-position is $(1, 3)$, whereas classical Wythoff has $(1, 2)$. Its exact P-positions are the strict $t = 2$ positions described below, with smaller-heap slope $\sqrt{2} \neq \varphi$ and larger-heap slope $\sqrt{2} + 2$. Thus the universal ratio-asymptotic claim fails under this convention too, already at $t = 1$.

The independent script `verify.py` directly computes the P-positions for $\leq$ at $t = 1$ on $[0, 100]^2$. It implements the non-strict rule directly and finds exact agreement with strict $t = 2$, and different positions from classical Wythoff. The first positions are $(0, 0), (1, 3), (2, 6), (4, 10), (5, 13)$; at index 29 the position $(41, 99)$ gives $a_{29}/29 \approx 1.413793103$. The exact integer formula at index

10^6 gives $a_n/n = 1.414213$. These finite checks are supplementary. Lean also proves the exact move and P-position equivalences and the resulting slope and refutation.

The words “with an explicit linear correction coefficient” do not specify a formula. They cannot preserve the stated asymptotic equivalence if the correction is of order n with nonzero coefficient. We address the alternative expansion reading explicitly in Section 5. We do not claim that an expansion allowing its leading slope to change is false.

2 A self-contained description of the P-positions for $t=2$

Put $r = \sqrt{2}$, $s = \sqrt{2} + 2$ and, for $n \in \mathbb{N}$, set

$$a_n = \lfloor nr \rfloor, \quad b_n = a_n + 2n = \lfloor ns \rfloor.$$

Both sequences start at zero and are strictly increasing, since $r > 1$ and $s > 1$. Also $b_n \geq a_n$, with equality only for $n = 0$.

Lemma 1 (Complementarity). *The positive terms of (a_n) and (b_n) partition the positive integers: each positive integer appears exactly once in exactly one of the two sequences.*

Proof. We have $r^2 = 2$ and

$$\frac{1}{r} + \frac{1}{s} = 1.$$

Both r and s are irrational. For each integer $N \geq 0$, the number of positive indices k for which $\lfloor kr \rfloor \leq N$ is $\lfloor (N+1)/r \rfloor$. Indeed, the condition is $k < (N+1)/r$, whose right side is irrational. The analogous count for s is $\lfloor (N+1)/s \rfloor$. Since these two irrational arguments sum to the integer $N+1$, their floors sum to N .

Thus the total number of appearances, counted with multiplicity, in $\{1, \dots, N\}$ is exactly N . Subtracting the count for $N-1$ shows that the integer $N \geq 1$ has exactly one appearance. No positive index contributes zero because both slopes exceed one. This proves both coverage and disjointness. Lean uses Mathlib’s proved Rayleigh–Beatty theorem for this step; its hypotheses are checked algebraically. $\square$

Call a position a candidate if it is (a_n, b_n) or (b_n, a_n) for some $n \geq 0$.

Lemma 2 (No move between candidates). *A legal 2-Wythoff move cannot connect two candidate positions.*

Proof. A move on one heap leaves the other coordinate unchanged. Strict increase implies that equal coordinates within the same sequence have equal indices; complementarity implies that equality between an a -coordinate and a b -coordinate is possible only at index zero. In either case a strict one-heap reduction between candidates is impossible.

For a move on both heaps, let the reductions be $k, l > 0$. Between candidates with the same orientation, $k - l$ equals $2(n - m)$ or its negative. It can have absolute value less than 2 only if $n = m$, which gives no strict reduction. Between opposite orientations, $k - l$ equals $2(n + m)$ or its negative. Its absolute value can be less than 2 only if $n = m = 0$, again with no strict reduction. Hence the required move is impossible. $\square$

Lemma 3 (Every other position can reach a candidate). *Every noncandidate position has a legal move to a candidate.*

Proof. By symmetry suppose $x \leq y$. Coverage gives either $x = a_n$ or $x = b_n$.

If $x = b_n$, then $a_n \leq b_n \leq y$. If $y = a_n$, the position is already the reversed candidate, contrary to the hypothesis. Thus $y > a_n$, and reducing the second heap gives (b_n, a_n) . This also covers $n = 0$ and $y > 0$.

If $x = a_n$, equality $y = b_n$ is excluded. If $y > b_n$, reduce the second heap to b_n . Otherwise $a_n \leq y < b_n = a_n + 2n$. Let

$$d = y - a_n, \quad m = \lfloor d/2 \rfloor.$$

Then $m < n$ and $a_m < a_n$. Since $2m \leq d$, we have $b_m = a_m + 2m < a_n + d = y$. Remove

$$k = a_n - a_m > 0, \quad l = y - b_m > 0.$$

Their difference is $l - k = d - 2m \in \{0, 1\}$, so $|k - l| < 2$. This is a legal move to (a_m, b_m) . $\square$

Theorem 1 (Exact game characterization). *The P-positions of 2-Wythoff are precisely the candidates. Consequently the sorted P-positions are exactly (a_n, b_n) in increasing order of a_n .*

Proof. Use strong induction on $x + y$. If a position is a candidate, all its followers are noncandidates by the no-move lemma; by induction they are not P-positions. Thus it is a P-position. If a position is not a candidate, it has a candidate follower, which is a P-position by induction; hence it is not a P-position. Strict increase gives the asserted enumeration and its uniqueness. $\square$

This also recovers the mex description, although a mex theorem is not needed in the Lean proof. The number a_n is absent from all earlier coordinates by strict increase and complementarity. Every $x < a_n$ appears as some a_i or b_i ; in either case $a_i \leq x < a_n$ implies $i < n$. Thus

$$a_n = \text{mex}\{a_i, b_i : i < n\}, \quad b_n = a_n + 2n.$$

3 Limit and contradiction

For $n > 0$ the floor inequalities give

$$0 \leq nr - a_n < 1, \quad 0 \leq r - \frac{a_n}{n} < \frac{1}{n}.$$

Since $1/n \rightarrow 0$, the squeeze theorem yields $a_n/n \rightarrow r$. Therefore

$$\frac{a_n}{\varphi n} \rightarrow \frac{r}{\varphi}.$$

This limit is not one. Indeed, $r^2 = 2$, whereas $\varphi^2 = \varphi + 1$; equality $r = \varphi$ would imply $r = 1$, contradicting $r > 1$. Uniqueness of real limits rules out the conjectured convergence to one. A universal claim over $t \geq 1$ implies its $t = 2$ instance, so that claim is refuted.

The same formulas give $b_n/n \rightarrow \sqrt{2} + 2$. Thus the conjecture's second leading slope also fails at $t = 2$.

4 Formal statement and semantic correspondence

The file `lean4/Wythoff.lean` defines `Move t p q` by the exact one-heap and two-heap move conditions. The two strict inequalities on the reductions encode $|k - l| < t$ without using signed subtraction. It defines `isP t p` recursively by

`isP t p = (forall q, Move t p q -> Not (isP t q))`

with termination measured by the sum of the coordinates. The terminating function `decideP` supplies a decision procedure by finite search over all smaller follower heaps.

The theorem `p_characterization` identifies this recursively defined P-predicate with the candidate set. The theorem `sorted_p_positions` characterizes all pairs with the smaller coordinate first. `a_strict` proves that their smaller heaps strictly increase.

We define `lower t x` to mean that there exists $y \geq x$ for which (x, y) is a P-position, and `nthLower t n := Nat.nth (lower t) n`. Thus `nthLower` refers directly to the game, not to an arbitrarily chosen Beatty sequence. `nthLower_two` proves that its $t = 2$ value equals `a n`. Infinitude and strict increase justify the use of `Nat.nth` without its finite-set fallback.

The main theorem has this Lean statement (ASCII transcription of notation):

```
theorem t_two_refutation :
  Not (Tendsto
    (fun n => (nthLower 2 n : Real) /
      (Real.goldenRatio * n))
    atTop (nhds 1))
```

The universal consequence is:

```
theorem universal_refutation :
  Not (forall t : Nat, 1 <= t ->
    Tendsto
      (fun n => (nthLower t n : Real) /
        (Real.goldenRatio * n))
      atTop (nhds 1))
```

A coordinatewise pair-asymptotic assertion entails this first-coordinate condition, so refuting the necessary condition refutes the full pair assertion. No assertion about the correction coefficient is substituted for the stated leading-term condition.

For the alternative convention, `MoveLE` implements the non-strict inequalities and `isPLE` defines the P-predicate by the same well-founded normal-play recursion. `moveLE_iff_move_succ` proves the exact move equivalence; `isPLE_iff_isP_succ` proves its consequence for all P-positions, by induction on total heap size. `nthLowerLE` enumerates the smaller heaps directly from this alternative game's P-predicate. `nthLowerLE_eq_succ` identifies that enumeration with the strict game's at parameter $t + 1$. `lowerLE_one_slope` proves its $\sqrt{2}$ limit at $t = 1$; `t_one_le_refutation` and `universal_le_refutation` refute the ratio assertion for that instance and universally, respectively.

5 Correction wording and other parameters

If the intended assertion instead allows an expansion with a leading-order correction, then the actual $t = 2$ result is

$$(a_n, b_n) = (\varphi n, (\varphi + 2)n) + (\sqrt{2} - \varphi)n(1, 1) + O(1).$$

The same floor error appears in both coordinates and has absolute value less than one. Lean also proves `correction_slope`, namely

$$\frac{a_n - \varphi n}{n} \longrightarrow \sqrt{2} - \varphi.$$

This is a true correction expansion. Its nonzero linear term changes the slope and therefore does not establish the literal asymptotic equivalence in the conjecture. A reading which permits that change is not refuted by this submission.

For context, Fraenkel’s general result gives

$$\alpha_t = \frac{2 - t + \sqrt{t^2 + 4}}{2}, \quad \beta_t = \alpha_t + t, \quad (a_n, b_n) = (\lfloor n\alpha_t \rfloor, \lfloor n\beta_t \rfloor).$$

It gives $\alpha_1 = \varphi$, so the stated leading asymptotics hold for $t = 1$. They fail for every integer $t \geq 2$: the relation $1/\alpha_t + 1/(\alpha_t + t) = 1$ is equivalent to $\alpha_t^2 + (t - 2)\alpha_t - t = 0$. If $\alpha_t = \varphi$, using $\varphi^2 = \varphi + 1$ gives $(t - 1)(\varphi - 1) = 0$, impossible for $t \geq 2$. This general observation uses the cited theorem and is not claimed as an additional Lean result; the fully formalized $t = 2$ case suffices for the disproof.

6 Verification and dependencies

The project pins Lean `leanprover/lean4:v4.33.0` and Mathlib `v4.33.0`. The Mathlib revision is

`db584cd6d46c92f209a44c0f1c829460d327499d`.

In an independent checkout, run `lake exe cache get && lake build` from `lean4/`. Compilation against this pinned Mathlib environment succeeded; the recorded compiler output and axiom audits are in `verification.txt`.

The main theorem audits at the end of the Lean file list exactly these standard axioms:

`[propext, Classical.choice, Quot.sound]`.

The proof uses no additional axioms, admissions, or native decision procedure. Mathlib’s Beatty theorem and its analytic limit lemmas are proved library results.

The independent standard-library Python script `verify.py` checks every heap pair in $[0, 100]^2$ using the legal moves and the normal-play recursion. It verifies exact agreement with the closed form using the integer identity $\lfloor n\sqrt{2} \rfloor = \text{isqrt}(2n^2)$, and checks the first 101 mex-generated pairs. It directly checks the non-strict $t = 1$ game against both strict games $t = 1$ and $t = 2$, lists its initial P-positions, computes its box-derived slope, and prints large-index ratios using the verified exact formula. These finite checks are supplementary; the Lean proof establishes the infinite claims without relying on Python.

References

- [1] A. S. Fraenkel, “How to beat your Wythoff games’ opponent on three fronts,” *The American Mathematical Monthly* **89** (1982), 353–361.
- [2] Mathlib contributors, Rayleigh’s theorem on Beatty sequences, Mathlib v4.33.0, file `Mathlib/NumberTheory/Rayleigh.lean`.

Attribution. Submitted by AlyciaBHZ on behalf of the Omega Institute (trureturing project, <https://github.com/the-omega-institute/trureturing>). No trureturing code is vendored.